

Year 10 Science (Adjusted)
Semester 1 Examination Revision 2019
Physical Sciences

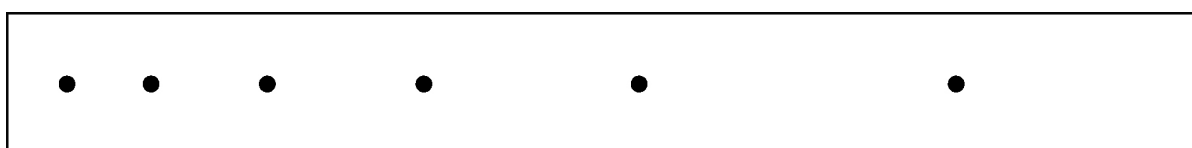
1. What is the difference between ***distance*** and ***displacement***? Draw a diagram to help your explanation.

2. Complete the following table. Make sure you include the appropriate units for each quantity.

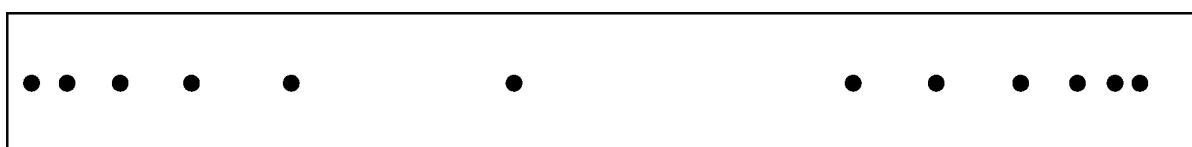
Distance travelled	Time taken	Average speed
400 m	45 s	
3000 km		600 km/h
	25 s	30 m/s
200 m		25 m/s

3. For each ticker tape shown, describe the motion of the body to which the tape was attached. The left-hand dot is at time zero.

(a)



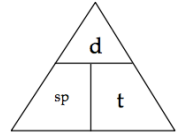
(b)



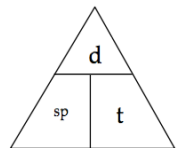
4. A man drives his car 800 m due East and then turns North, driving 300 m before stopping at a petrol station. It took him 120 s to get there.

(a) What distance he has driven.

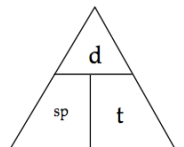
(b) Calculate his average speed for the journey.



5. A balloon filled with hydrogen gas has a velocity of 40 m/s East. How long will it take to go 450 m ?

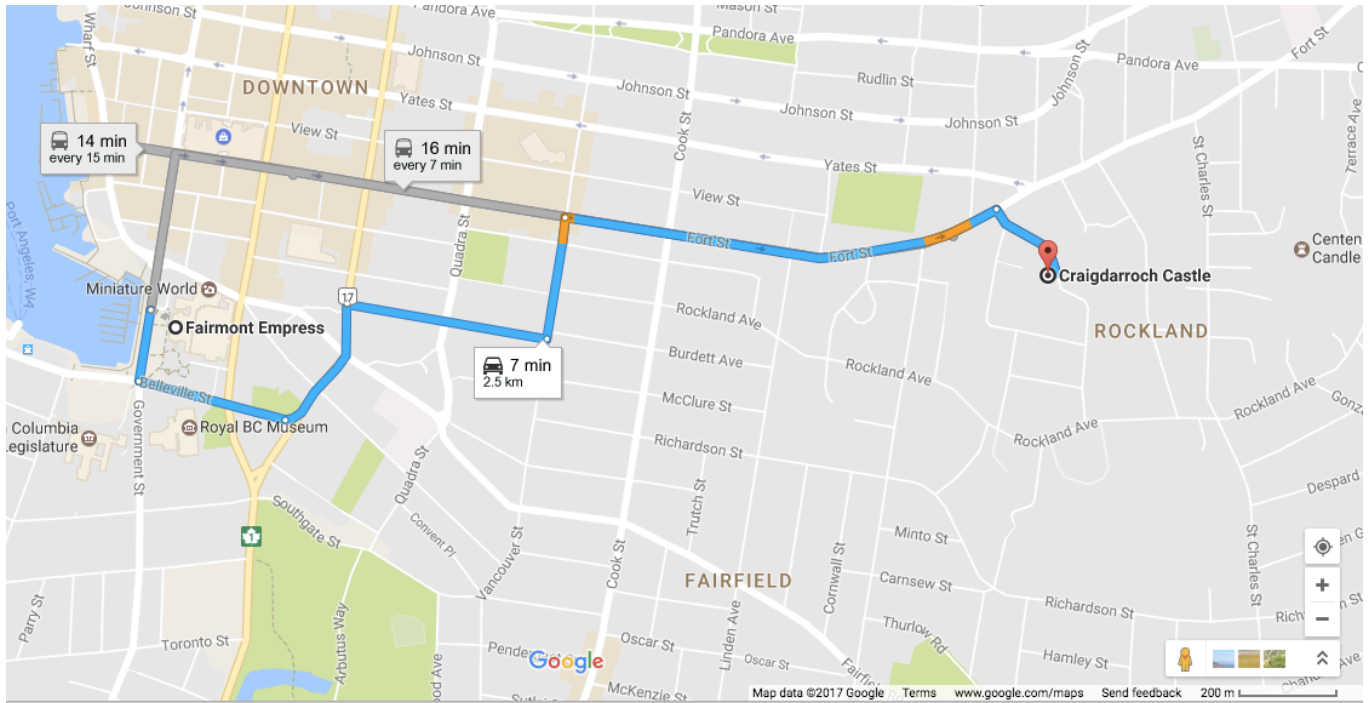


6. What is the average speed of a plane if it flies 2700 km in 2.3 hours? Give the answer in km/h.



7. Use the scale in the map below to work out the **distance and displacement** from the Fairmont Empress hotel to Craigdarroch Castle. Use either the blue or grey route.

Remember to give a bearing or direction for the displacement.



distance: _____

displacement: _____